XUFEI LIU

The Wharton School, University of Pennsylvania
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EDUCATION

University of Pennsylvania, The Wharton School, Philadelphia, PA 2022 – 2027 (expected)
Ph.D. Candidate in Operations Management, expected graduation May 2027
Advisors: Gad Allon and Ken Moon
Dissertation committee: Gad Allon, Ken Moon, Hummy Song, Christian Terwiesch

Georgia Institute of Technology, Atlanta, GA 2019 – 2022
B.S. in Industrial and Systems Engineering, May 2022
Honors Program; GPA 4.0/4.0

RESEARCH INTERESTS

Human-AI Collaboration; AI; Machine Learning; Healthcare Operations; Education; Adaptive Learning Operations; Worker Wellbeing and Fatigue

JOB MARKET PAPER

The First Draft Problem: Productivity, Quality, and Physiological Strain in LLM-Assisted Writing
with Hummy Song and Christian Terwiesch

WORKS IN PROGRESS

When to Step In: Decision-Aware Representation Learning for Real-Time AI Interventions
with Gad Allon and Ken Moon
Status: In preparation for submission

Learning How to Learn: Adaptive Algorithms for Long-Term Memory
with Gad Allon and Ken Moon
Status: Field experiment in progress

Supervision Cost: Trust, Fatigue, and Error Detection When Editing LLM Drafts
with Hummy Song and Christian Terwiesch
Status: Data collection in progress

Targeting Knowledge Networks through LLM-Enabled Personalization
Status: Study design in progress

OTHER PUBLICATIONS

Harrington, J., Hua, X., **Liu, X.**, Nash, A., Rios, R., and Wong, T. W. H. (2023). "Probabilistic chip-collecting games with modulo winning conditions." *Discrete Applied Mathematics*, 324, 93–98.

Glassband, J., Koch, G., Lebiere, S., **Liu, X.**, and Sabini, E. (2021). "On the Assignment Graphs of Oriented Graphs." arXiv:2111.04882.

Fiorini, E., Glassband, J., Koch, G., Lebiere, S., **Liu, X.**, Sabini, E., Shank, N. B., and Woldar, A. (2021). Four sequences published in the On-Line Encyclopedia of Integer Sequences (A346189, A346198, A346199, A346204).

HONORS & AWARDS

Mack Institute Research Fellowship ($13,000, $9,500, $9,000) 2023, 2025, 2026
Fishman-Davidson Center Grant ($5,000) 2026
Wharton AI and Analytics Initiative ($12,000) 2026
George James Award, The Wharton School ($600, $800) 2024, 2025

TEACHING

University of Pennsylvania, The Wharton School 2023 – present

Instructor, Wharton Doctoral Math Camp Summer 2023, 2024

Sole instructor of the mathematics preparation course for incoming Wharton Ph.D. students (10 sessions per summer). Designed, created, and delivered the full syllabus covering real analysis, optimization, and probability.

MBA Program Teaching Assistant, Scaling Operations (OIDD 6360) Fall 2025

MBA elective taught by Gad Allon. Supported case discussions, held office hours, answered student questions, and graded assignments and exams.

CONFERENCE PRESENTATIONS

INFORMS Annual Meeting	2026
EMPOM: Workshop on Empirical Research in OM	2026
INFORMS Healthcare	2026
INFORMS Annual Meeting	2023, 2024, 2025
INFORMS MSOM Conference	2024, 2025
POMS Annual Conference	2025
Purdue Supply Chain and Operations Management Conference	2025
MIT Rising Scholars Conference	2023

SERVICE & LEADERSHIP

Organizer, Wharton–INSEAD Doctoral Consortium Fall 2024

- Served on a committee of six organizing the consortium; managed the paper review process and hosted two parallel tracks with 35 presenters, two panels, and two keynotes.

President and OIDD Representative, Wharton Doctoral Council 2022 – 2024

- Led the council representing the Wharton Ph.D. community; organized 31 events and managed a $32,000 budget.
- Worked with administration to create the annual Wharton GAPSA Fellowship.

Wharton Doctoral Representative, GAPSA and Penn Research Council 2022 – 2023

- Represented Wharton Ph.D. students in Penn's Graduate and Professional Student Assembly; allocated travel grant funding and served on the university Research Council.

SKILLS

Programming: Python, R, JavaScript, Qualtrics (custom JavaScript instrumentation)

Methods: Experimental design (lab and field), structural estimation, reinforcement learning and Markov decision processes, semi-supervised and decision-focused learning, optimization, causal inference (instrumental variables, difference-in-differences, regression discontinuity)

REFERENCES

Gad Allon (Advisor)
Professor, The Wharton School
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Ken Moon (Co-Advisor)
Associate Professor, Cornell SC Johnson
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